

1 **Hierarchical Vision–Language Navigation–Sign Understanding through**
2 **Correspondence-Aware Multiview Representation**

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1 ABSTRACT

2 **Objectives** Camera-based navigation in large transportation facilities requires interpreting signs despite small text,
3 perspective distortion, visual clutter, and repeated video observations. This study evaluates how spatial evidence con-
4 struction and temporal observation selection affect destination–direction reasoning.

5 **Methods** Navigation signs are localized, segmented, and projectively rectified, then represented as an annotated
6 scene plus all successfully rectified sign views linked by shared identifiers. A vision–language model interprets this
7 correspondence-aware input to produce constrained destination–direction labels. For video, optical character recogni-
8 tion binds queried destinations to tracked signs, and temporal policies select observations for multiview inference. We
9 evaluated seven fixed representation bundles on 346 images from more than ten campus buildings and 1,100 questions,
10 and temporal policies and frame strides on 25 videos and 133 questions.

11 **Findings** Full multiview answered 1,083/1,100 questions (98.45%), versus 1,065/1,100 (96.82%) for Original only, a
12 1.64-point gain that reduced errors from 35 to 17 (51.43%). The image-cluster bootstrap 95% confidence interval was
13 0.18–3.29 points, while exact cluster randomization gave $p = 0.0523$; only 18 images differed, so broad image-level
14 dominance was not established. In recorded video, Track-Lifetime Best View achieved the highest observed stride-6
15 accuracy (112/133, 84.21%); no policy difference was statistically supported. Stride 12 was computationally cheaper
16 than stride 3 (819 versus 3,236 processed frames) without a supported accuracy difference.

17 **Novelty** The principal contribution is a correspondence-aware multiview representation and its controlled evaluation
18 for navigation-sign understanding, treating the visual evidence supplied to the vision–language model (VLM) as a
19 design variable rather than a fixed observation.

20 **Practical Applications** The results inform the design of camera-based wayfinding tools for transit stations, airports,
21 campus facilities, and other large public buildings, including assistive navigation interfaces and robotic guides. Deploy-
22 ment would require site-specific validation, continuous-camera testing, and integration with route-level navigation.

1 INTRODUCTION

2 Airports, rail stations, transit hubs, and other large public buildings depend on visual wayfinding infrastructure. Navigation signs communicate destinations that may lie outside the current field of view and encode the local actions needed to reach them through combinations of text, arrows, pictograms, and spatial layout. Humans use this information to move through unfamiliar facilities without first constructing a complete map. Camera-based assistive systems and mobile robots could similarly use existing signage as human-designed, task-relevant spatial information.

7 Interpreting a navigation sign is more demanding than detecting a sign or recognizing its text. A useful system must identify the destination relevant to a query, associate it with the correct directional indicator, distinguish directional instructions from locational labels, and resolve signs containing multiple destinations or shared arrows. These decisions must often be made when the sign occupies a small portion of a cluttered scene, appears under perspective distortion, or is visible only briefly from a moving camera. Optical character recognition can recover portions of the text, but text alone does not preserve the visual relationships that determine which arrow applies to which destination.

13 Recent work formalized navigation-sign understanding as detecting signs and extracting structured destination–direction relationships with vision–language models (VLMs) (Agrawal et al. 2025). Subsequent work grounded sign-derived semantic instructions in local three-dimensional scene geometry for mapless robot navigation (Zimmerman et al. 2026). These studies establish the value of signage for robot autonomy, but they do not systematically determine how the evidence presented to the reasoning model should combine global scene context, localized sign detail, geometric normalization, and explicit correspondence. A single scene image preserves environmental context but may provide an oblique or low-resolution view of the sign. An isolated crop improves local visibility but discards the sign’s relationship to the surrounding scene, while an unlinked collection of crops leaves the model to infer which local view belongs to which region. In video, the system must additionally select among repeated, imperfect observations while avoiding unnecessary processing and VLM requests.

23 This paper presents a hierarchical framework for camera-based navigation-sign understanding. An open-vocabulary detector first localizes candidate signs. Segmentation and projective rectification then generate normalized sign views, while an annotated version of the original scene preserves the location of each candidate. These inputs are organized into a correspondence-aware multiview representation that explicitly links every rectified sign to its source region. The representation—rather than any individual detector, segmenter, or reasoning model—is the principal spatial contribution: it exposes global context, localized detail, and their correspondence in one structured VLM request. The VLM then answers destination-specific questions using a constrained direction vocabulary. For recorded video, text-based destination retrieval, persistent tracking, and temporal candidate selection provide one representative observation to the same multiview reasoning stage.

32 The evaluation follows a staged progression from spatial representation to temporal selection. On a frozen still-image benchmark of 346 images and 1,100 questions, the Full multiview representation achieved 98.45% accuracy, compared with 96.82% for the unmodified-image baseline. This 1.64-point gain reduced errors from 35 to 17 (51.43%); Full multiview corrected 27 baseline errors while introducing nine. The image-cluster bootstrap 95% confidence interval for the gain was 0.18–3.29 points, while exact cluster randomization gave $p = 0.0523$, providing qualified support for an aggregate improvement rather than broad image-by-image dominance. Recorded-video experiments on 25 videos and 133 questions compared four policies using identical stride-6 caches. Track-Lifetime Best View achieved the highest observed accuracy (112/133, 84.21%), but no policy difference was statistically supported. Across Track-Lifetime sampling conditions, strides 3 and 6 both answered 112/133 questions and stride 12 answered 111/133 while processing 819 rather than 3,236 frames. These results characterize temporal and sampling trade-offs rather than establish policy superiority.

43 The principal contributions of this work are:

- 44 • a correspondence-aware multiview representation that combines an annotated global scene with explicitly linked, perspective-rectified sign views for destination-specific navigation reasoning;
- 45 • a controlled evaluation of seven fixed representation bundles on 346 images and 1,100 questions, with paired, image-cluster-aware, and class-imbalance-sensitive analyses that quantify performance without claiming isolated component effects; and
- 46 • a recorded-video extension on 25 videos and 133 questions that compares temporal observation selection and source-frame sampling in terms of retrieval, navigation accuracy, selection delay, and preprocessing cost.

52 Together, the experiments examine not only whether a VLM can interpret navigation signs, but how its visual evidence should be constructed and, in a limited recorded-video setting, selected over time. Correct destination–direction association is a perception prerequisite for future camera-based navigation assistance and robotic wayfinding;

deployment, route-level decisions, and embodied navigation are outside the present scope.

RELATED WORK

Navigation Sign Understanding

Navigation-sign perception has traditionally been approached as detection or recognition. Unlike standardized traffic signs, indoor signs vary in appearance, placement, layout, and illumination and are frequently observed amid clutter or occlusion. Indoor-sign datasets and accessible-navigation systems therefore emphasized reliable localization (Almeida et al. 2019; Cheraghi et al. 2021), while later detectors separately localized arrows, icons, and text (De Vera et al. 2024). These methods provide useful perceptual primitives, but component detection alone does not establish which direction applies to a queried destination on a sign containing multiple destinations or shared arrows. This separates navigation-sign understanding from closed-set classification: the vocabulary is open, and the output depends on relations among elements rather than their identities alone.

Other work has treated signage as spatial knowledge. Recognized text has been incorporated into semantic maps (Case et al. 2011), used for localization (Zimmerman et al. 2022), and translated into local navigation actions (Liang et al. 2020). Chen et al. (2025) combined signage recognition with venue-map priors and exploration planning. Collectively, these systems establish the navigational value of signs but generally use text as a landmark or depend on task-specific parsing assumptions.

Agrawal et al. (2025) formalized *navigational-sign understanding* as sign detection followed by extraction of structured location–direction cues. Their diverse benchmark and open-set detection–VLM baseline demonstrate the promise of general-purpose multimodal reasoning while identifying failures involving small or oblique signs, dense layouts, multiline names, and ambiguous destination–arrow associations. Their robot deployment also uses active alignment to improve the observation. The work is therefore the closest formulation and baseline for our task: it shows both the feasibility of VLM parsing and its dependence on viewpoint and available evidence. More recently, SignScene grounded semantic instructions from signs in the local three-dimensional scene for mapless navigation (Zimmerman et al. 2026). This reinforces that spatial representation affects sign grounding, but neither work systematically isolates how global and local image evidence, geometric normalization, and explicit view correspondence affect destination–direction interpretation. We address that gap with a correspondence-aware multiview representation and a controlled visual-input ablation. The next subsection places this task within broader VLM-based robotic navigation.

Vision–Language Models for Robotic Navigation

Large pretrained models have expanded robotic navigation beyond fixed object categories and narrowly specified goals. LM-Nav combines pretrained language, vision, and control models for long-horizon natural-language navigation (Shah et al. 2023), while RoboHop organizes observations into a topological representation for open-world semantic navigation (Garg et al. 2024). Navigation signs align naturally with language-specified goals, but interpreting them additionally requires recovering the structured relationship between a destination and its directional indicator. These works establish the value of language-aligned evidence for navigation but do not focus on extracting instructions from visually structured signs.

Open-vocabulary spatial representations also demonstrate how multimodal features can support perception and planning. ConceptGraphs associates language-aligned objects with a three-dimensional scene graph (Gu et al. 2024), whereas signs provide complementary top-down knowledge about destinations outside the current field of view. Exploiting that knowledge first requires reliable interpretation of fine-grained sign content and layout. These representations motivate the downstream value of sign-derived relations without prescribing how the sign observation itself should be constructed.

General-purpose VLMs such as Gemini jointly process visual and textual inputs and produce structured responses (Gemini Team 2023), reflecting the broader shift toward adaptable multimodal systems (J. Zhang et al. 2024). Yet their reasoning remains constrained by the evidence supplied at inference: a distant scene may obscure sign content, while an isolated crop discards context and an unlinked collection of crops obscures correspondence. VLM-navigation research largely emphasizes goal interpretation, semantic reasoning, and planning while assuming an adequate observation is available. Our still-image ablation instead holds the VLM and prompt fixed and varies how global context, localized detail, normalization, and correspondence are organized. This motivates examining the perception and temporal-selection stages that construct those observations.

Visual Representation and Temporal Observation

Open-vocabulary detectors such as Grounding DINO and YOLO-World localize objects specified through language rather than fixed trained categories (Liu et al. 2024; Cheng et al. 2024). OCR systems detect and recognize text in natural images (Du et al. 2020; Liao et al. 2020), but transcription omits arrows, pictograms, grouping, and layout. Promptable segmentation with MobileSAM can refine a detection into an estimated sign surface suitable for geometric normalization (C. Zhang et al. 2023). These tools address complementary perception problems, yet prior pipelines typically evaluate them independently or pass one resulting crop downstream rather than explicitly linking scene, raw-crop, and normalized views. Localized image-description research likewise shows the value of retaining fine regional detail together with broader context, although it addresses region captioning rather than navigation-sign relations (Lian et al. 2025). The representation question is therefore distinct from component accuracy: the same detections and crops may support different reasoning outcomes depending on whether their context and correspondence are retained.

Video adds repeated observations with changing scale, sharpness, occlusion, and completeness. ByteTrack associates detections across frames and maintains object identities despite confidence variation (Y. Zhang et al. 2022), but tracking does not decide which observation is most useful for VLM inference. Generic video work has selected a representative, high-quality frame (Ren et al. 2020) or adaptively sampled frames for efficient recognition (Wu et al. 2019). Query-aware selection has also been studied for video-language reasoning, where the question guides which frames are passed to a downstream model (Buch et al. 2025). Navigation-sign understanding instead requires an observation that exposes fine text and the complete destination–direction relationship. Active sign-reading systems may improve the viewpoint through robot motion, but recorded video instead requires choosing among observations already captured.

Prior work therefore supplies effective detections, text, masks, tracks, and general frame-selection principles, but does not evaluate synchronized global–local sign representations or destination-conditioned selection over a sign track. Our correspondence-aware multiview representation submits an annotated scene with explicitly linked rectified views and ablates alternative organizations. Track-Lifetime Best View binds a retrieved destination to a track and retains its highest-ranked observation until that track leaves the scene. Together, they treat representation and temporal observation as controlled design variables rather than fixed preprocessing assumptions.

METHODOLOGY

Problem Formulation

We formulate navigation-sign understanding as a visual question answering task. Given an input image containing zero or more navigation signs and a set of navigation-related questions, the objective is to predict the appropriate navigation direction for each queried destination by reasoning over sign content and layout rather than only detecting signs or transcribing text. The system must associate the destination with its directional indicator, distinguish guidance from locational labels, and handle multiple destinations or shared arrows.

Formally, let $I \in \mathbb{R}^{H \times W \times 3}$ denote an RGB scene image and let

$$Q = \{q_1, q_2, \dots, q_n\} \quad (1)$$

denote the set of navigation questions associated with that image. Each question q_i requests the direction of a particular destination, such as “Where is the Men’s Restroom?” or “Where is Platform 2?” The framework predicts an answer $\hat{y}_i$ from a constrained navigation vocabulary $\mathcal{D}$:

$$\mathcal{D} = \{\text{left}, \text{right}, \text{straight}, \text{straight-left}, \text{straight-right}, \\ \text{down}, \text{down-left}, \text{down-right}, \text{locational}, \text{u-turn}, \\ \text{unknown}, \text{uncertain}\}. \quad (2)$$

Multiple labels may be returned when more than one direction is valid. The desired mapping can therefore be written as

$$f(I, Q) = \{\hat{y}_1, \hat{y}_2, \dots, \hat{y}_n\}, \quad \hat{y}_i \in \mathcal{D} \text{ or } \hat{y}_i \subseteq \mathcal{D}. \quad (3)$$

1 Framework Overview

2 Figure 1 summarizes the proposed hierarchy. An open-vocabulary detector localizes candidate signs, which are segmented and rectified; an annotated scene preserves correspondence between each region and its derived local views.

3 The annotated scene and synchronized rectified views form a correspondence-aware multiview representation
4 that a vision–language model (VLM) interprets to return a structured direction for each question, separating evidence
5 construction from semantic reasoning. The hierarchy thus exposes task-relevant evidence before semantic association
6 and constrained answer generation.
7

8 Open-Vocabulary Navigation-Sign Localization

9 The framework uses YOLO-World for open-vocabulary localization. The still-image implementation prompts
10 `yolo_v8s-world.pt` with “navigation sign,” “directional sign,” “wayfinding sign,” “sign with arrow,” “directional
11 arrow sign,” “hallway directional sign,” and “exit sign,” without retraining for each design.

12 For each input image, the detector returns candidate sign regions and confidence scores. Highly overlapping
13 detections are filtered using an intersection-over-union criterion, and the retained regions are forwarded to the subse-
14 quent segmentation and rectification stage. Each retained region is also assigned an identifier that links its location in
15 the annotated scene to its corresponding local sign view within the same multiview request.

16 Sign-Surface Segmentation and Perspective Rectification

17 Each detected region is geometrically normalized before semantic interpretation. MobileSAM-vit_t, prompted by
18 the detector box, returns multiple masks; the implementation retains the mask with the highest predicted score and
19 extracts its largest external contour. Douglas–Peucker polygon approximation uses $\epsilon = 0.02P$, where P is the closed-
20 contour perimeter. An approximation with exactly four vertices supplies the sign corners; otherwise, the four corners
21 of the contour’s minimum-area rectangle are used. The ordered corners define a projective warp whose width and
22 height are the larger lengths of each pair of opposite sides. A detection produces no rectified view if no contour exists
23 or the resulting width or height is nonpositive. Both still-image preprocessing and recorded-video post-selection call
24 this same rectification implementation. The resulting view is retained with the annotated scene rather than interpreted
25 independently.

26 Correspondence-Aware Multiview Representation

27 An original scene preserves context but may show a sign at small scale, whereas a local crop enlarges its content while
28 removing that context. The framework constructs a correspondence-aware multiview representation consisting of an
29 annotated original scene and perspective-rectified views of successfully processed navigation signs. The annotated
30 scene serves as a contextual anchor, while each rectified view provides a normalized close-up of one candidate sign.
31 A shared identifier explicitly links every local view to its source region in the original scene. Figure 2 illustrates the
32 S1 and S2 correspondences within a scene containing multiple detected regions.

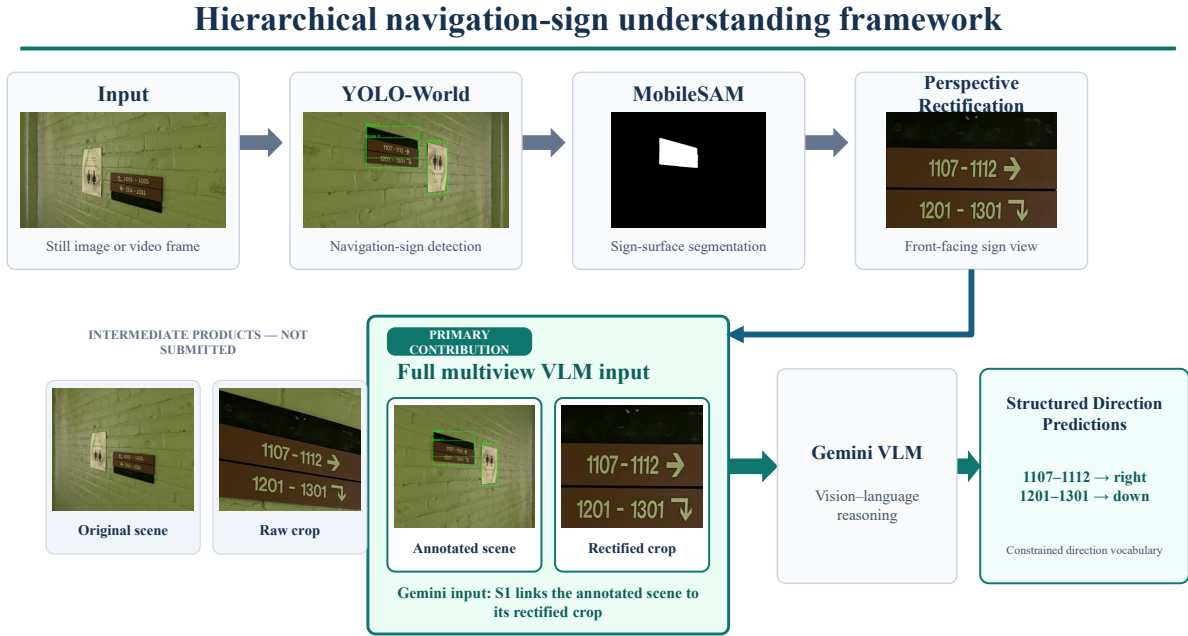


FIGURE 1 Overview of the proposed hierarchical navigation-sign understanding framework. An input scene image or recorded-video frame is processed by YOLO-World for navigation-sign detection, MobileSAM for sign-surface segmentation, and projective rectification for geometric normalization. The unannotated scene and raw sign crops are intermediate products; the annotated scene and its explicitly corresponding rectified crops form the multiview input interpreted by the Gemini vision-language model to produce structured direction predictions.

The complete representation is submitted in one VLM request so that the overview locates each sign while the rectified views expose its detailed content. Shared identifiers allow the model to associate each local view with its region in the overview. The representation thereby preserves environmental relationships without requiring small sign details to remain legible in the full-resolution scene.

Vision-Language Navigation Reasoning

The VLM jointly interprets the annotated scene and its associated rectified sign views. For each image, it receives the predefined navigation questions associated with that benchmark item. The model must identify the sign relevant to each question, associate the queried destination with the appropriate directional indicator, and return the corresponding navigation label.

The framework is conceptually independent of a particular VLM, provided that the model can jointly reason over multiple image inputs and produce a constrained structured response. The current implementation uses Gemini as the multimodal reasoning component.

Structured Answer Generation

The final stage returns structured navigation answers rather than free-form natural language. Each output entry directly associates a navigation question with its predicted direction from $\mathcal{D}$. The constrained response format reduces ambiguity, simplifies downstream use by a navigation system, and enables automatic comparison with benchmark annotations. These structured predictions constitute the final output evaluated in the Experimental Evaluation section.

Correspondence-aware multiview representation

Shared identifiers link each detected region to its corresponding rectified sign view

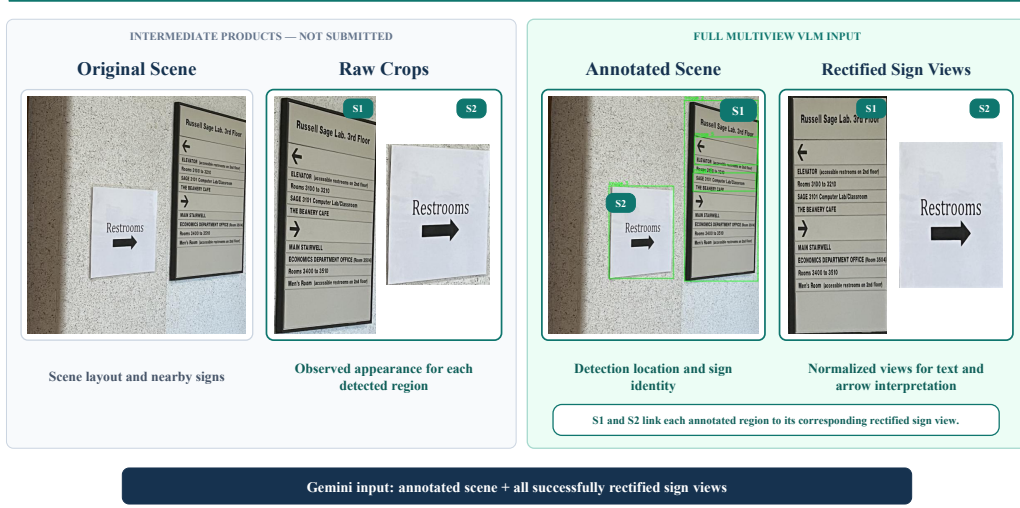


FIGURE 2 Correspondence-aware multiview representation. The original scene and raw crops are intermediate products and are not submitted. Shared identifiers S1 and S2 illustrate how each detected region in the annotated scene maps to its corresponding rectified sign view. The complete Full multiview input comprises the annotated scene and every successfully generated rectified sign view.

1 Reference Semantics and Evaluation Scoring

2 Benchmark references use three explicit semantic types. A *scalar* reference requires the deduplicated predicted set
3 to equal its one-label set; additional labels are incorrect. An *alternatives* reference lists admissible coarse or fine
4 descriptions of one direction, and a prediction is correct when its deduplicated set is nonempty and contains only
5 admissible labels. A *joint-required* reference encodes multiple directions that must all be returned, so the predicted
6 and required sets must be equal. Ordering and duplicate labels do not affect any comparison.

7 The operational scorer existed before inference, and the benchmark manifest was frozen before the still-
8 image runs. A pre-submission semantic audit subsequently identified inconsistent treatment of multi-label references.
9 The author adjudicated the ambiguous annotations, after which the stored predictions were rescored under the three
10 reference types without new VLM calls. The original annotation and inference artifacts remain preserved; the corrected
11 references and derived summaries are versioned separately.

12 The expanded video audit applied the same semantics: legacy scalar references were interpreted as scalar,
13 whereas list-valued references required an explicit type. Independent rescoring corrected four stored legacy correctness
14 flags associated with IMG_3769.MOV, question_0000, without changing its stored predictions, caches, or annotations.

15 Recorded-Video Observation Retrieval and Temporal Selection

16 For recorded video, the still-image pipeline is extended with temporal association, destination retrieval, and observation
17 selection before multiview inference. ByteTrack associates sign detections across processed frames, and the pipeline’s
18 stable-identifier assignment forms persistent sign tracks. OCR is applied to each detected crop, and RapidFuzz partial-
19 ratio matching compares the recognized text with the destination extracted from each question. A qualifying match
20 retrieves a candidate observation for that destination.

21 Track-Lifetime Best View permanently binds the destination to the stable track of its first qualifying OCR
22 match and never transfers the binding to another track. While the bound track remains active, the method retains its
23 strongest visual observation using a quality score that combines crop area, Laplacian sharpness, detector confidence,
24 and an edge-cutoff penalty. OCR evidence is retained separately, allowing a later visual observation to replace the
25 initial matched view. Specifically,

$$q_{\text{visual}} = A \left(1 + \frac{\min(L, 500)}{500} \right) (0.5 + \text{clip}(c_{\text{det}}, 0, 1)) p_e,$$

$$p_e = \begin{cases} 1.00, & n_e = 0, \\ 0.65, & n_e = 1, \\ 0.40, & n_e \geq 2, \end{cases} \quad (4)$$

where A is crop area in pixels, L is the variance of the 64-bit Laplacian of the grayscale crop, and c_{det} is detector confidence. An edge touch is counted for each side of the detection box lying within four pixels of the corresponding frame boundary, and n_e is the number of touched sides. There are no additional weights or normalization terms. A track’s best visual view is replaced only by a strictly larger q_{visual} ; ties retain the earlier view, irrespective of OCR confidence or match score. Qualifying OCR evidence is maintained separately and is replaced only when $(\bar{c}_{\text{OCR}}, s_{\text{match}})$ is lexicographically greater; an exact tie retains the earlier OCR evidence. After each observation of the bound track, the method forms a candidate by pairing the retained best visual frame and detection with the separately retained OCR text, confidence, and match score. Thus, an OCR improvement can update the evidence associated with a candidate but cannot change the retained visual frame unless q_{visual} also increases. Complete candidates are ranked as

$$r_{\text{TLBV}} = (q_{\text{visual}}, \bar{c}_{\text{OCR}}, s_{\text{match}}), \quad (5)$$

where q_{visual} is visual quality, $\bar{c}_{\text{OCR}}$ is mean OCR confidence, and s_{match} is the fuzzy destination-match score; higher tuples are preferred, and an equal tuple retains the earlier candidate. The selection window closes after the bound track remains absent for eight consecutive processed updates and is force-closed at video end. The retained source frame and detection are then segmented and rectified. The annotated frame and its explicitly corresponding rectified sign view form the same multiview representation used for still images and are submitted with the destination question in one VLM request.

EXPERIMENTAL EVALUATION

The evaluation progressed from still-image representation ablations to recorded-video observation selection and frame sampling. This sequence separates spatial and temporal effects.

Experimental Setup

Dataset

We collected a still-image benchmark consisting of 346 navigation sign images across more than ten buildings on the Rensselaer Polytechnic Institute (RPI) campus. Images were captured using an Apple iPhone 13 while traversing academic, administrative, and public campus buildings containing indoor navigation signage. Table 1 summarizes the benchmark, and Figure 3 illustrates representative examples.

TABLE 1 Summary of the still-image benchmark.

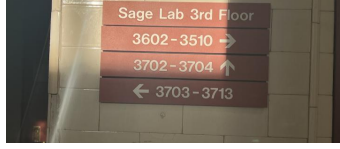
Statistic	Value
Images	346
Destination-oriented QA pairs	1,100
Collection location	RPI campus
Buildings surveyed	>10
Camera	Apple iPhone 13
Image resolution	4032×3024 pixels

Representative examples from the still-image benchmark

Variation in sign geometry, density, scene context, and illumination



IMG_3300 - Oblique directory with multiple adjacent signs



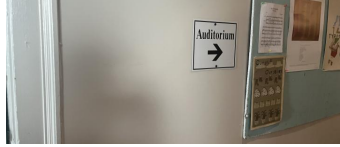
IMG_3305 - Glare across a dense multi-arrow directory



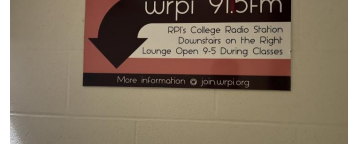
IMG_3347 - Compact sign under strong perspective distortion



IMG_3391 - Cluttered stairwell with multiple small signs



IMG_3396 - Sparse pictogram-dominant directional layout



IMG_3464 - Nonstandard text-and-graphics directional sign



IMG_3550 - Mixed sign scales amid strong visual distractors



IMG_3646 - Many directories and plaques in one scene



IMG_3669 - Backlit corridor with uneven illumination

FIGURE 3 Representative examples from the still-image benchmark. The scenes span compact and directory-style layouts, frontal and oblique views, single- and multiple-sign settings, visual clutter, glare, backlighting, and uneven illumination.

Rather than collecting images under controlled laboratory conditions, the dataset was intentionally designed to reflect realistic pedestrian viewpoints. Images were captured from frontal and oblique viewing angles, at varying distances, and under diverse environmental conditions. Consequently, the benchmark contains challenging scenarios including glare, motion blur, varying illumination, overexposed scenes, partial occlusions, and multiple navigation signs appearing within a single image.

For evaluation, a total of 1,100 destination-oriented visual question answering (VQA) pairs were manually constructed. Each question follows the form “Where is *destination*?” where the destination corresponds to a navigational location appearing on the sign. Synonymous destination names (e.g., “Bathroom” and “Restroom”) were included when appropriate to better reflect natural user queries. Ground-truth navigation directions were verified by inspecting each navigation sign to ensure accurate correspondence between destinations and directional indicators.

Evaluation Task and Metrics

The system receives an image or a selected video frame together with one or more destination-specific questions, such as “Where is Restrooms?” For each question, the VLM predicts a navigation direction. The permitted output vocabulary includes left, right, straight, straight-left, straight-right, down, down-left, down-right, locational, u-turn, unknown, and uncertain. A list may be returned when multiple directions are valid.

Predictions were evaluated at the question level using the scalar, alternatives, and joint-required semantics defined in the Reference Semantics and Evaluation Scoring subsection. Question-answering accuracy was calculated as

$$\text{Accuracy} = \frac{N_{\text{correct}}}{N_{\text{evaluated}}}. \quad (6)$$

Preprocessing failures remained in the evaluation denominator and were counted as incorrect. The reported accuracy therefore measures end-to-end question answering rather than accuracy conditional on successful sign detection or crop generation.

Class-sensitive analyses used mutually exclusive exact reference-semantics categories and mutually exclusive higher-level direction families. Complete multi-label reference sets were retained rather than assigned to one constituent direction. We report micro and unweighted macro accuracy, with a sensitivity analysis restricted to categories or families containing at least ten questions. Family-specific image-cluster bootstrap intervals were exploratory and unadjusted for multiple comparisons.

Cluster-Aware Statistical Inference

Because questions from the same source image may be dependent, the paired still-image comparison was also evaluated with image-cluster-aware analyses. The primary cluster bootstrap sampled 346 complete source-image clusters with replacement and retained all questions belonging to each sampled image. It used 100,000 replicates and fixed random seed 20260723. A paired exact randomization test swapped the two representation labels for all questions within each image, while a macro analysis gave every image equal weight. Clustering by image does not create independence and cannot account for different images showing the same physical sign because no verified physical-sign mapping exists.

Still-Image Ablation

Objective

The still-image ablation evaluated how different visual representations of the same navigation-sign scene affected direction-question answering. The image benchmark, navigation questions, detector outputs, rectification outputs, prompt, and VLM configuration were held fixed. Only the composition and organization of the visual inputs supplied to the VLM varied.

The ablation tested whether local views were independently sufficient and whether combining context, explicit localization, and rectified evidence improved navigation understanding.

Benchmark Controls

The benchmark and question ordering described in the Experimental Setup subsection were frozen across conditions and verified using image hashes and the benchmark manifest.

The benchmark contained ten direction labels: left, right, straight, straight-left, straight-right, down, down-left, down-right, locational, and u-turn. The corrected reference manifest contains 1,092 scalar questions, three alternative-answer questions, and five joint-required questions. Although unknown and uncertain were available model outputs, neither appeared as a ground-truth class in the still-image benchmark.

Visual Representations

Seven visual-input conditions were evaluated:

1. **Original only:** the unmodified scene without detection annotations, crops, or rectification.
2. **Raw crops only:** detected sign crops without the original scene or rectification.
3. **Rectified crops only:** rectified sign crops without the original scene.
4. **Original + raw crops:** the unmodified scene followed by raw crops, without annotations linking crops to regions in the scene.
5. **Original + rectified crops:** the unmodified scene followed by rectified crops, without explicit correspondence annotations.
6. **Annotated original + raw crops:** the original scene with numbered detection boxes followed by the corresponding raw crops.
7. **Full multiview:** the annotated original scene with numbered detection boxes followed by the corresponding rectified close-ups.

In the Full multiview condition, the prompt explicitly associated each rectified view with its numbered region in the annotated scene. This condition represents the complete proposed spatial hierarchy: the original scene preserves global context, detection identifies candidate navigation signs, and rectification provides localized views for detailed interpretation.

All crop-based conditions used a shared frozen preprocessing cache. Sign candidates were generated using YOLO-World with `yolo8s-world.pt`, a detection-confidence threshold of 0.05, and an intersection-over-union threshold of 0.6 for duplicate removal. Rectified views were generated using MobileSAM with the `vit_t` architecture. Reusing the same cached outputs ensured that differences among conditions resulted from the representation supplied to the VLM rather than changes in detection or segmentation outputs.

This bundle-level ablation is not a complete factorial experiment: moving from Original only to a multiview

condition jointly changes localization, magnification, annotation, and image count. Original + rectified crops versus Full multiview jointly changes scene annotation and correspondence, so those factors cannot be separated causally. The fixed bundles therefore do not independently identify component or interaction effects.

VLM Configuration

All conditions used gemini-3.5-flash with temperature set to zero, one candidate response, and high image detail. Voting, PaddleOCR, and tracking were disabled. All questions associated with one image were submitted together in a single logical request. Requests affected by API or response-format failures were retried until a valid structured response was obtained. Because recorded request latency includes uncontrolled API retries and randomized retry delays, it was not treated as a controlled comparison of computational efficiency.

Quantitative Results

Table 2 reports end-to-end question accuracy for all seven representations. Full multiview produced the highest observed accuracy, correctly answering 1,083 of 1,100 questions (98.45%). This was an absolute improvement of 1.64 percentage points over Original only, which correctly answered 1,065 questions (96.82%). Full multiview therefore reduced the error count from 35 to 17—18 fewer errors, corresponding to a 51.43% relative error reduction.

Because every representation was evaluated on the same questions, Full multiview and Original only were also compared using paired outcomes (Table 3). Full multiview corrected 27 questions missed by Original only while introducing nine errors on questions answered correctly by the baseline. The conditions agreed on the remaining 1,064 questions: both were correct on 1,056 and both were incorrect on eight.

A question-level two-sided exact McNemar test over the 36 discordant pairs produced $p = 0.0039$. Because multiple questions can originate from the same image, this test is treated as supporting evidence rather than a complete account of statistical uncertainty.

The observed Full-minus-Original micro-accuracy difference was 1.636 percentage points. In the image-cluster bootstrap, its 95% percentile confidence interval was 0.181 to 3.293 points, whereas the 99% interval was -0.209 to 3.900 points. Thus, the primary 95% interval excluded zero but the 99% interval did not. The exact image-cluster randomization test gave $p = 0.0523$. Giving every image equal weight yielded macro accuracies of 97.4029% for Original only and 98.3510% for Full multiview, a 0.9481-point difference whose 95% cluster bootstrap interval was -0.6463 to 2.5683 points. Full multiview had higher image-level accuracy on 11 images, Original only on seven, and the conditions tied on 328; the exact sign test over the 18 non-tied images gave $p = 0.4807$.

Horizontal directions comprised 908/1,100 questions (82.55%). Among exact categories with at least ten questions, unweighted macro accuracy increased from 93.77% for Original only to 97.06% for Full multiview; among supported direction families, it increased from 95.92% to 97.38%. The net 18-question improvement primarily reflected gains for *right* (402/422 to 413/422) and *down* (17/25 to 23/25). Performance declined slightly for *left* (481/482 to 480/482) and *straight* (59/59 to 57/59), while diagonal and locational family accuracy was unchanged. Four net corrections came from the eight-question u-turn category, which is too small to support generalization. Thus, Full multiview remained favorable under both supported macro summaries, but the class-level effects were heterogeneous.

To contextualize the primary Original-only versus Full-multiview comparison, an exploratory all-pairs analysis examined the 21 representation contrasts. Full multiview ranked first in micro, supported exact-category macro, supported direction-family macro, and image-level macro accuracy. Versus Annotated original + raw crops, it improved 42 questions and regressed on five (net +37); the cluster-bootstrap 95% interval was 0.74 to 7.27 points. Eighteen of the 42 improvements occurred in one image, and the three most-improved images accounted for 57.14% of all 42 improvements; the cluster-randomization contrast did not survive Holm correction. Versus Original + rectified crops, it improved 35 and regressed on four (net +31); the interval was -0.28 to 7.12 points, again without adjusted cluster-randomization support. After Holm correction, only Full multiview versus Raw crops only and Rectified crops only retained cluster-randomization support.

Discussion

Crop-only, unlinked-view, and annotated-raw configurations had lower observed aggregate accuracy. Full multiview was the best observed bundle under micro, class-macro, family-macro, and image-macro summaries. Selected contrasts were consistent with rectification benefiting the annotated-scene bundle, but gains were concentrated and lacked multiplicity-adjusted cluster-randomization support; most pairwise differences were not distinguishable after clustering and correction. Because the seven conditions are fixed bundles, they do not independently isolate annotation, correspondence, rectification, localization, image count, or their interactions.

TABLE 2 Still-image visual-representation ablation. Preprocessing failures remained in the 1,100-question denominator and were counted as incorrect.

Visual representation	Correct/Total	Accuracy	Failed images
Original only	1,065/1,100	96.82%	0
Raw crops only	1,030/1,100	93.64%	4
Rectified crops only	1,023/1,100	93.00%	4
Original + raw crops	1,058/1,100	96.18%	0
Original + rectified crops	1,052/1,100	95.64%	0
Annotated original + raw crops	1,046/1,100	95.09%	0
Full multiview	1,083/1,100	98.45%	0

TABLE 3 Paired question outcomes for Original only and Full multiview.

Paired outcome	Questions
Both correct	1,056
Original wrong, Full multiview correct	27
Original correct, Full multiview wrong	9
Both wrong	8

The clustered results provide qualified support for an aggregate question-level improvement rather than consistent image-by-image dominance: the net gain was concentrated among the 18 images with different image-level accuracies. Repeated photographs of the same physical sign may introduce additional dependence that image-level clustering cannot address.

The 98.45% micro accuracy is influenced by the benchmark’s 82.55% horizontal share. Favorable supported-category and supported-family macro comparisons indicate that the aggregate conclusion is not solely an artifact of micro weighting, but category-level effects were heterogeneous: right and down improved, left and straight declined slightly, and the other supported families were unchanged. All exploratory family-specific image-cluster 95% intervals included zero and were not adjusted for multiple comparisons; they therefore do not support claims of reliable family-level superiority. Rare categories remain too small for generalization.

Qualitative Analysis

For IMG_3347.JPG, Original only returned down-right for rooms 1201–1301, whereas Full multiview returned the reference direction, down, consistent with the clearer orientation in the rectified view. The example does not isolate causality and also exposed imperfect duplicate suppression: overlapping detections produced redundant views, including one low-confidence detection.

For IMG_3391.JPG, Full multiview returned straight-right rather than straight for two floor questions, consistent with interference from a nearby sign containing a curved up-right arrow. Detection and rectification therefore do not guarantee correct association across adjacent signs.

For IMG_3464.JPG, the reference down follows the word “Downstairs,” while the predicted down-left follows the curved arrow; the sign also states “on the Right.” Exact-label evaluation counted the plausible alternative as incorrect, and two formulations of the destination produced two question errors from this single visual ambiguity.

Based on the quantitative results, Full multiview was carried forward into the recorded-video pipeline. Each selected video observation was represented by an annotated overview and corresponding rectified sign views, allowing the video experiments to focus on temporal observation selection rather than reopening the spatial-representation question.

Recorded-Video Experiments

Objective and Benchmark

The recorded-video evaluation extended the still-image framework to sequences containing repeated observations of navigation signs. The study addressed two questions: how the policy used to select a representative observation affects navigation-question accuracy and selection delay, and how source-frame sampling density trades off destination retrieval, question accuracy, and preprocessing cost. Temporal information was used for retrieval and observation

selection; multiple frames were not jointly supplied to the VLM.

The combined benchmark contained 25 videos and 133 questions: the frozen original set contributed 13 videos and 78 questions, and an extension contributed 12 videos and 55 questions. Of the 133 questions, 132 referred to destinations expected to appear and one was expected absent. The annotations contained 93 exact destination strings and used down, left, locational, right, straight, straight-left, straight-right, and unknown. The frozen cache metadata recorded 9,684 source frames across the combined videos.

The number of unique physical signs cannot be determined from the benchmark. Multiple questions can refer to destinations on one sign, and algorithmic track identifiers do not constitute physical-sign ground truth. In particular, IMG_3857.MOV and IMG_3858.MOV are distinct files with identical question sets; without verified physical-sign identities, they cannot be treated as independent sign instances. The repository also does not record the camera model, acquisition protocol, camera height, walking speed, stabilization settings, or whether any videos were repeated takes.

Observation Cache and Destination Retrieval

The recorded-video methodology was evaluated from cached observations. At source-frame stride 6, processed frames were passed to YOLO-World using yolov8s-world.pt. The detector used a confidence threshold of 0.05, an intersection-over-union threshold of 0.3, and class-agnostic non-maximum suppression. ByteTrack, stable-identifier assignment, and PaddleOCR with English recognition and angle classification were applied once to construct the cache. The video detector used the seven text prompts listed in the methodology plus an empty-string prompt, as recorded in the frozen cache configuration. This differs from still-image preprocessing, which applied order-preserving, class-independent duplicate removal at intersection-over-union greater than 0.6 rather than the video’s class-agnostic non-maximum suppression.

For each question, the destination string was extracted from the question text and matched against recognized text using RapidFuzz partial_ratio. A candidate qualified when its match score was at least 70. When a destination contained digits, every numeric target token was also required to appear exactly among the recognized numeric tokens. This prevented a fuzzy match between distinct room or platform numbers.

For Fixed Window and Last Seen, qualifying observations were ranked using

$$s_{\text{candidate}} = s_{\text{match}} + 10\bar{c}_{\text{OCR}}, \quad (7)$$

where s_{match} is the fuzzy destination-match score and $\bar{c}_{\text{OCR}}$ is the mean OCR confidence. A strictly greater score replaced the current best observation, while a tie retained the earlier candidate. First Match selected the first qualifying observation without subsequent score optimization.

Track-Lifetime Best View used the track binding and candidate rank in Eq. 5; its visual-quality values were read from the cache rather than recomputed during policy evaluation.

For a successfully retrieved destination, the selected observation was converted to the correspondence-aware multiview input described in the recorded-video methodology. If no candidate qualified, rectification and VLM inference were skipped and the prediction defaulted to unknown.

Retrieval therefore denotes the presence of at least one automatically matched OCR candidate with score at least 70; it does not indicate that a human verified the correct physical sign. An unretrieved present destination was counted as incorrect, whereas an unretrieved expected-absent destination was correctly scored as unknown.

Temporal Observation Policies

Four temporal policies were compared using identical stride-6 observation caches:

- **First Match** selected the first chronologically qualifying detection without subsequent score optimization.
- **Fixed Window** began at the first qualifying processed frame i , considered qualifying candidates through processed frame position $i + 5$ inclusive, and selected the highest-ranked candidate. The implemented window could therefore contain the initial position and five later processed positions. At stride 6 and approximately 30 frames/s, it spanned at most approximately one second of source video.
- **Last Seen** bound selection to the stable track identifier of the first qualifying candidate, continued updating the best candidate on that track, and stopped after eight consecutive processed updates without observing the bound track.
- **Track-Lifetime Best View** also bound the destination to the stable track of the first qualifying match, but continuously retained the highest-quality visual observation seen over that track’s lifetime. It reused

qualifying OCR evidence from the bound track when a stronger visual observation became available and closed after eight consecutive processed updates without the track or at video end. Figure 4 qualitatively compares the observation selected by the first three policies with the later observation retained by Track-Lifetime Best View for a representative navigation question.

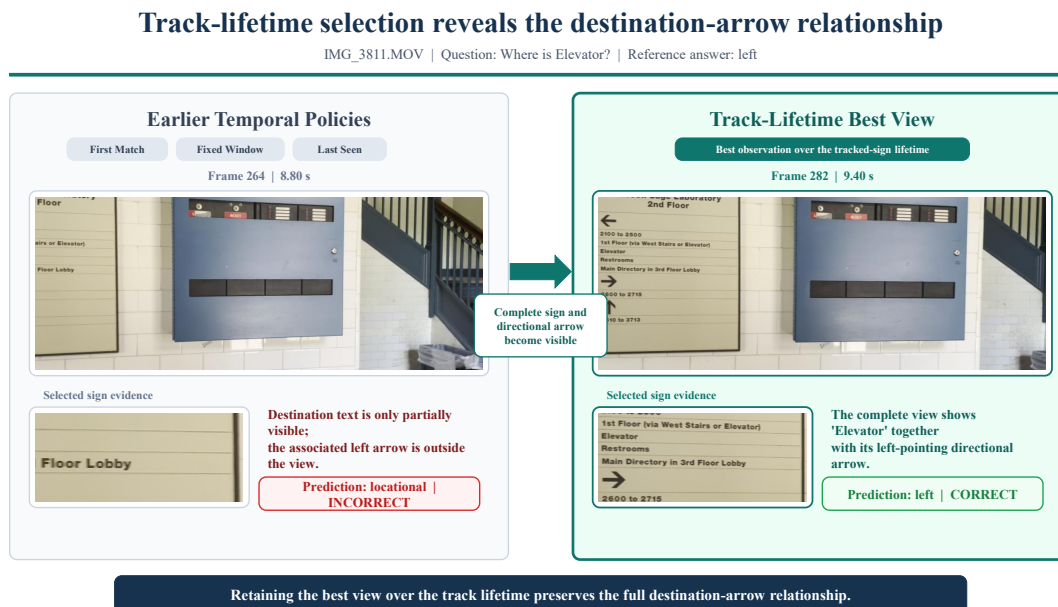


FIGURE 4 Temporal selections for the Elevator query in video IMG_3811.MOV. The first three policies chose frame 264, whose local view omitted the associated arrow, and predicted locational. Track-Lifetime Best View retained frame 282, where the destination and left-pointing arrow appear together, and correctly predicted left.

The completed matrix contains 150 per-video condition records. Of these, 98 persist gemini-3.5-flash; 52 legacy records from the original stride-6 runs do not persist a model identifier. Their exact model therefore cannot be independently confirmed from stored outputs and is not inferred from the current runner default.

All four policies reused the same frozen detections, stable tracks, OCR outputs, destination matches, prompt, and evaluation procedure. The fourth policy consumed visual-quality values already stored in the cache and did not rerun YOLO, tracking, OCR, or destination matching. VLM outputs were not shared across policies; each retrieved policy-question pair invoked the multiview backend independently, even when two policies selected the same source frame.

Table 4 reports the independently rescored combined results. Track-Lifetime Best View had the highest observed accuracy, 112/133 (84.21%), compared with 111/133 for Last Seen and 110/133 for both First Match and Fixed Window; all policies retrieved 124/133 destinations. Mean and median delays are paired to First Match over those 124 questions. Four Track-Lifetime selections have a delay of -0.2 s because a later OCR match can bind a track whose retained best visual observation occurred earlier.

The observed policy differences were small and concentrated. All four policies were correct on 108 questions and incorrect on 20; Track-Lifetime alone corrected one question and introduced no unique regression. Relative to Last Seen, 24 videos tied and only IMG_3811.MOV favored Track-Lifetime. No question-level, video-cluster, bootstrap, or multiplicity-adjusted comparison supported a reliable policy difference; all six Holm-adjusted policy-comparison p -values were 1.000. Figure 4 illustrates the single Track-Lifetime-only correction without implying a general mechanism.

TABLE 4 Recorded-video temporal-policy ablation at source-frame stride 6. Delay is the selected source-video timestamp minus the First Match timestamp over 124 questions with paired valid timestamps; the nine unretrieved questions have no selection timestamp and are excluded.

Policy	Correct	Accuracy	Retrieved	Mean delay	Median delay
First Match	110/133	82.71%	124/133	0.000 s	0.000 s
Fixed Window	110/133	82.71%	124/133	0.297 s	0.200 s
Last Seen	111/133	83.46%	124/133	0.361 s	0.000 s
Track-Lifetime Best View	112/133	84.21%	124/133	0.519 s	0.100 s

1 Source-Frame Stride Ablation

2 The source-frame stride ablation used Track-Lifetime Best View and independently constructed observation caches at
3 strides 3, 6, and 12. These settings processed source-frame indices 0, 3, 6, . . . ; 0, 6, 12, . . . ; and 0, 12, 24, . . . , respec-
4 tively. Because each stride used a separate cache, detection, tracking, OCR, and destination matching were rerun at
5 each sampling density. The stride-6 Track-Lifetime results were shared with the temporal-policy comparison rather
6 than recomputed.

TABLE 5 Recorded-video source-frame stride ablation using Track-Lifetime Best View. Cache runtimes are single observed preprocessing runs and exclude frame selection, rectification, and VLM inference.

Stride	Correct	Accuracy	Retrieved	Processed frames	Source frames processed	Cache runtime
3	112/133	84.21%	127/133	3,236	33.42%	1,510.754 s
6	112/133	84.21%	124/133	1,625	16.78%	1,092.222 s
12	111/133	83.46%	127/133	819	8.46%	664.968 s

7 As shown in Table 5, strides 3 and 6 tied at 112/133 (84.21%), while stride 12 answered 111/133 (83.46%).
8 Stride 12 processed 819 rather than 3,236 frames, a 74.69% reduction, and its recorded cache runtime was 55.99%
9 lower than stride 3. No accuracy difference was statistically supported: every stride McNemar, cluster-randomization,
10 and Holm-adjusted comparison gave $p = 1.000$. Correctness changed on 19 questions across the three Track-Lifetime
11 conditions, so the near-equal totals do not imply identical behavior.

12 The runtime results should also be interpreted as observed end-to-end cache generation times rather than con-
13 trolled hardware benchmarks. Timing included video decoding, YOLO/tracking, crop generation, OCR, destination
14 matching, and cache construction, but excluded subsequent selection, rectification, and VLM inference. Each value
15 represents one recorded run; hardware specifications, warm-up behavior, background workload, and repeated-run vari-
16 ability were not recorded.

17 Recorded-Video Discussion

18 Temporal-policy accuracy ranged from 82.71% to 84.21% at the same 124/133 retrieval count. Track-Lifetime Best
19 View had the highest observed accuracy, but its gain was concentrated and no comparison supported a reliable differ-
20 ence. The original/extension split provides a sensitivity view rather than independent confirmation: stride-6 Track-
21 Lifetime answered 64/78 original and 48/55 extension questions (112/133 combined), while on the extension it tied
22 First Match and Last Seen. The extension therefore preserved the highest combined Track-Lifetime total but weakened
23 the apparent policy advantage.

24 Coarser sampling substantially reduced processed frames and recorded cache runtime without a supported
25 accuracy difference. On the extension alone, stride 3 answered 50/55 questions, whereas strides 6 and 12 each answered
26 48/55. These split results are descriptive, and the runtime values are single observed cache-generation runs rather than
27 controlled hardware benchmarks.

28 Several limitations constrain generalization. The benchmark contains 25 videos, one question is expected ab-
29 sent, and retrieval is defined by automatic OCR matching rather than human verification of the selected physical sign.
30 Multiple questions can also refer to the same sign, so question outcomes are not necessarily independent; repeated
31 recordings may add dependence that cannot be resolved without verified physical-sign mappings. Finally, the timing
32 data omit VLM latency and were not collected through repeated controlled runs. The recorded-video results there-
33 fore characterize offline temporal-selection and sampling trade-offs. Continuous live-camera evaluation of end-to-end
34 latency, throughput, queue behavior, memory use, and tracker stability remains an important direction for future work.

CONCLUSION

Navigation-sign understanding requires more than detecting a sign or transcribing its text: the reasoning model must preserve the relationship between a queried destination, its directional indicator, and the surrounding scene. This study addressed that problem through a hierarchical framework that constructs an annotated global scene, produces perspective-rectified local sign views, and explicitly links each local view to its source region. The resulting correspondence-aware multiview representation gives the VLM both contextual and localized evidence without requiring it to infer which crop belongs to which detected sign.

On the frozen still-image benchmark, Full multiview was the best observed representation bundle, answering 1,083 of 1,100 questions correctly (98.45%), compared with 1,065 (96.82%) for Original only. This 1.64-point improvement reduced errors from 35 to 17, a 51.43% relative reduction; Full multiview corrected 27 Original-only errors while introducing nine. The 95% image-cluster bootstrap interval supported an aggregate gain, whereas exact cluster randomization gave $p = 0.0523$ and only 18 images differed in image-level accuracy. The result therefore remained favorable under question-level and supported-category summaries without establishing consistent image-by-image or direction-class superiority. Because the seven conditions were fixed representation bundles, they also did not independently identify the causal contribution of annotation, correspondence, rectification, localization, image count, or their interactions.

The recorded-video experiments extended the study from spatial representation to temporal observation selection. Track-Lifetime Best View achieved the highest observed stride-6 result, 112 of 133 questions (84.21%), at the same 124/133 retrieval count as the other policies. No policy difference was statistically supported, and relative to Last Seen, 24 videos tied and only IMG_3811.MOV favored Track-Lifetime. The source-frame ablation showed that stride 12 processed 74.69% fewer frames than stride 3 and had 55.99% lower recorded cache runtime, while strides 3 and 6 answered 112/133 questions and stride 12 answered 111/133; no accuracy difference was supported. These results motivate larger prospective video evaluation rather than identify a universally optimal policy or sampling rate.

Several limitations constrain generalization. Data were collected on one campus, direction classes were strongly imbalanced, and multiple questions from the same image or physical sign make question outcomes nonindependent. The recorded-video benchmark contains only 25 videos, and retrieval was defined by automatic OCR matching rather than human verification of the selected physical sign. Repeated videos may show the same physical sign, but no verified mapping exists to model that dependence. Cache runtimes were single observed runs, and their timing excluded later selection, rectification, and VLM inference. Finally, the study did not evaluate continuous live-camera operation, mobile-robot integration, or route-level navigation.

Future work should evaluate more balanced, multi-site datasets and larger recorded-video benchmarks. Continuous live-camera feasibility testing should measure end-to-end latency, throughput, queue and memory behavior, and tracking stability. A separate embodied extension could then integrate the perception framework with a mobile robot and evaluate closed-loop, route-level navigation. These are distinct evaluation stages: live-camera testing assesses continuous perception, whereas robot experiments assess navigation behavior. Within the present offline scope, the study identifies correspondence-aware Full multiview as the best evaluated representation bundle. It also shows that the visual evidence supplied to a VLM is an important perception design variable: preserving destination-direction relationships is a prerequisite for future camera-based navigation assistance and robotic wayfinding, although deployment and route-level performance remain to be demonstrated.

GENERATIVE AI DISCLOSURE

OpenAI ChatGPT and Codex assisted with portions of code development and verification, manuscript organization and language editing, reference compilation and formatting, and figure preparation. Anthropic Claude assisted with manuscript critique and replacement wording during revision; it did not generate experimental data or independently conduct experiments. The authors reviewed the resulting materials and retain responsibility for the experimental procedures, results, citations, figures, manuscript text, and the accuracy and integrity of the work. The Gemini vision-language model evaluated as part of the proposed system is a component of the reported research methodology and is distinct from the generative-AI assistance used to prepare the manuscript.

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